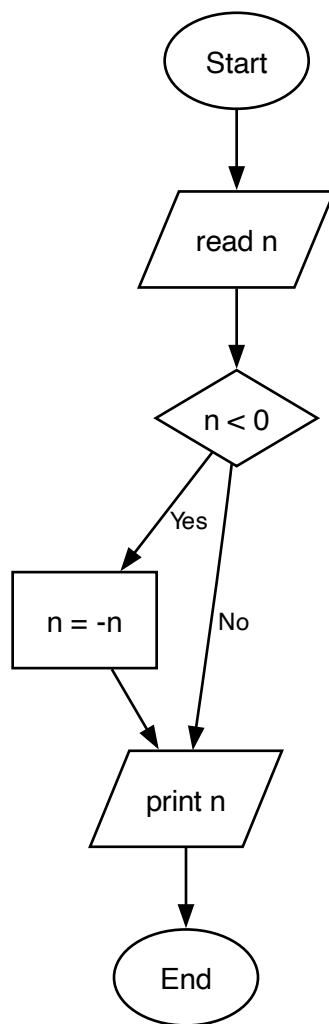


Flowchart Worksheet

Answer key

Problem 1.



(i) For each input below, follow the flowchart and write the output.

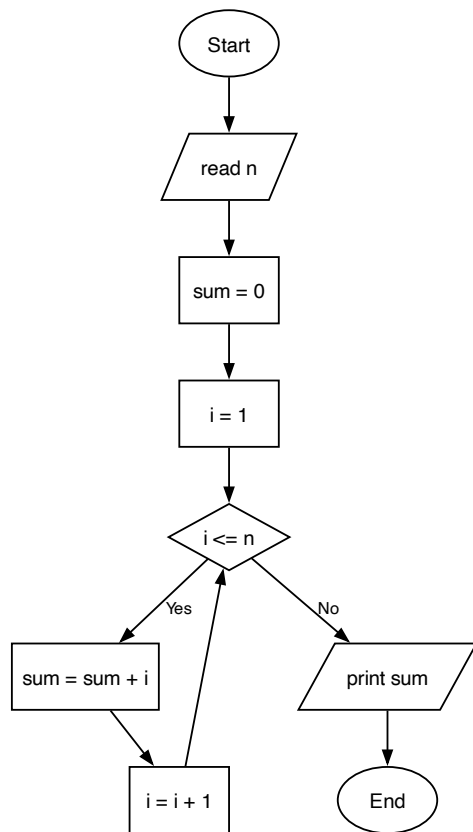
input n	output
7	7
-4	4
0	0
-19	19

(ii) Can you tell what this algorithm is doing?

Answer: Absolute value

Problem 2.

Trace this algorithm for $n = 4$, and complete the trace table.

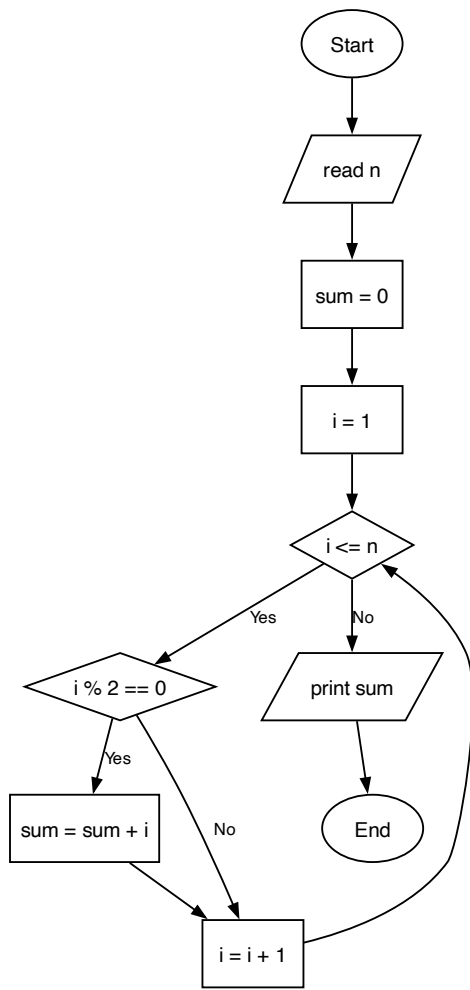


n	sum	i	output
4			
4	0		
4	0	1	
4	1	1	
4	1	2	
4	3	2	
4	3	3	
4	6	3	
4	6	4	
4	10	4	
4	10	5	
4	10	5	10

Output: 10

Problem 3.

Trace this algorithm for $n = 5$, and complete the trace table.



n	sum	i	output
5			
5	0		
5	0	1	
5	0	2	
5	2	2	
5	2	3	
5	2	4	
5	6	4	
5	6	5	
5	6	6	
5	6	6	6

Output: 6